



Full length article

Phase evolution in two-phase alloys during severe plastic deformation

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ABSTRACT

Phase evolution in FCC metals with strongly interacting alloy components during severe plastic deformation is investigated using molecular dynamics simulations. Specifically, we study the alloy microstructure in steady state, nucleation and growth of precipitates in supersaturated alloys, and the dissolution of precipitates in undersaturated alloys. The results are compared to a modified effective temperature model, providing a physical understanding for the atomic processes underlying the model and a perspective on its strengths and weaknesses. Key observations in this work are nucleation and growth of precipitates during SPD at a temperature of 100 K; Gibbs-Thomson-like behavior relating steady-state solubility to precipitate size under sustained shearing; a direct relationship between the effective temperature and the shear modulus; and the importance of cluster agglomeration during precipitate growth. The study also reveals that the mechanism of forced chemical mixing depends on precipitate size, adding complications for effective temperature models describing inhomogeneous systems. The simulations are shown to provide good semi-quantitative agreement with experimental findings reported in the literature.

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1. Introduction

Severe plastic deformation (SPD) of metal alloys at low temperatures generally leads to forced chemical mixing and homogenization of alloying constituents, commonly referred to as mechanical alloying [1]. A surprising finding about this behavior is that even at very low temperatures, far too low for vacancy mediated diffusion processes, alloys with large enthalpies of mixing, either positive or negative, show phase separation or chemical order after extended deformation [2–5]. Many of these systems, moreover, tend to approach these steady-state microstructures independent of their initial conditions [6–9], making SPD an attractive processing method for developing advanced nanocomposite alloys with high strength, microstructural stability at high temperatures, good creep resistance, high electrical/thermal conductivity, and resistance to radiation damage [10–13]. Notably, similar behaviors have been reported in high heat-of-mixing systems for the related process of energetic particle irradiation [14–17].

While there have been no general theories developed specifically for SPD, an appealing model often used to describe driven alloys [4,18–20] is the effective temperature model (ETM) proposed by Martin [21], as the steady-state microstructure during SPD in this model corresponds simply to the equilibrium microstructure

at some higher (effective) temperature. In the original model, however, the effective temperature is given by the expression,

$$T_{eff} = T(1 + \gamma_b^\infty) \quad (1)$$

where $\gamma_b^\infty = D_b^\infty/D_{th}$ represents the ratio of diffusion coefficients arising from ballistic (assumed random) atomic jumps, D_b^∞ , and thermally activated atomic jumps (usually mediated by point defects), D_{th} . At low temperature $D_{th} \rightarrow 0$ and $T_{eff} \rightarrow \infty$, and all alloys are predicted to homogenize. This ETM model thus fails to describe phase separation in highly immiscible alloys during low temperature SPD. In this light, it should be noted that “low temperature” for SPD in many engineering alloys can be room temperature (RT), as diffusion is mediated by vacancies, and vacancies are virtually immobile in many metals at 300 K. The Martin model, however, was proposed initially to describe materials subjected to irradiation with energetic particles, and ballistic jumps referred to energetic recoil events, typically on the order of 10 eV or more, far greater than chemical binding energies. SPD, in contrast, involves local shear stresses, a few GPa at most, that force atoms over local potential barriers, thus providing an energy of < 1 eV per atom. Consequently, strong interatomic interactions are likely to bias this motion and give rise to short-range ordering. Although such chemically-biased “ballistic” mixing processes were not explicitly included in the original ETM, it is straightforward to do so, as recently demonstrated by Verma et al. [8]. In that work, the infinite temperature ballistic diffusion coefficient in Martin’s model (Eq. (1)), D_b^∞ , was replaced with a new diffusion coefficient

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cient, $D_b^{T_0}$, to convey the non-random nature of displacements due to the applied shearing with some characteristic process temperature, T_0 , that is now finite. Thus, the total diffusional flux could be expressed as

$$J_{tot} = -M_{th}(T) \cdot \nabla \mu(T) - M_b(T_0) \cdot \nabla \mu(T_0) \\ = -c\bar{c} \left\{ \frac{D_{th}(T)}{kT} \nabla \mu(T) + \frac{D_b(T_0)}{kT_0} \nabla \mu(T_0) \right\} \quad (2)$$

where M_{th} and M_b are, respectively, the mobilities for thermally-activated and ballistic transport; c is the solute concentration and $\bar{c} = (1 - c)$; $D_{th}(T)$ is the thermal diffusion coefficient (including any enhancement by shear-induced point defects); $D_b(T_0)$ is the 'ballistic', shear-mediated diffusion coefficient with a characteristic temperature T_0 ; finally, $\mu(T)$ and $\mu(T_0)$ are the chemical potentials at temperature T and T_0 . Note that $D_b(T_0)$ is simply the diffusion coefficient arising from forced chemical mixing; it is weakly dependent on temperature, hence we will use the notation $D_b(T_0) = D_b^{T_0}$ hereinafter. By assuming a regular solution model to evaluate the chemical potentials in Eq. (2), an *effective potential* for the driven system can be formed, in direct correspondence to the method employed in ref [21], and it now yields an effective temperature given by

$$T_{eff} = T(1 + \gamma_b^{T_0}) / \left(1 + \frac{T}{T_0} \gamma_b^{T_0}\right), \quad (3)$$

where the forcing intensity is $\gamma_b^{T_0} = D_b^{T_0}/D_{th}$ (see ref. [8] for additional details). Recall that in the effective temperature model, the steady state of a system driven at a given temperature, T , corresponds to the equilibrium state of the system at some higher temperature given by T_{eff} . Thus, as $T \rightarrow 0$ in this model, the effective temperature tends to T_0 and not to infinity. An METM proposed earlier by Lund and Schuh yields a similar result [22]. We can roughly estimate the magnitude of T_0 in the present model by noting that the maximum work performed per unit volume during shear deformation is $\sim \tau_{th} \cdot \epsilon$, where τ_{th} is the theoretical shear strength of the material and ϵ is the strain. For Cu, assuming $\tau_{th} \approx G/10$ (where G is the shear modulus) [23], $\tau_{th}^{Cu} \approx 4.5$ GPa, so that the average work done on each atom undergoing one jump during shear, i.e., after a local strain of unity, is ~ 0.35 eV yielding $T_0 \sim 4,000$ K. Corresponding behavior during particle irradiation involves thermal spikes, whereby diffusion takes place in a local liquid-like environment so that in Cu, $T_0 \sim 2,000$ K [24], explaining why phase evolution during SPD and irradiation are often quite similar and why neither leads to homogenization in strongly immiscible alloys.

While the above METM can rationalize why highly immiscible alloys show only limited solubilities during low temperature SPD, there has been rather little consideration of this model in more quantitative terms. For example, can an effective free energy of the alloy during low-temperature SPD be represented by the true free energy at T_0 , or how does T_0 depend on alloy properties and the details of the forced atomic transport mechanisms. The implications of this model, moreover, for the kinetics of phase evolution during SPD have also not been explored. These same questions arose in early studies of phase evolution during energetic particle irradiation [25], but with the help of molecular dynamics simulation [15,26–28], they are now largely answered, at least on a semi-quantitative basis. In the present work we follow this same approach and apply molecular dynamics simulations to explore these questions for SPD, examining not only the alloy phases present in steady state during SPD, but also nucleation and growth of precipitates in supersaturated alloys, and precipitate dissolution in under-saturated alloys. Key observations of this work in evaluating the ef-

fective temperature model are precipitate nucleation and coarsening during low-temperature SPD, as well as Gibbs-Thomson (G-T) like behavior. A second important finding establishes a connection between the effective temperature, T_0 , and the elastic properties of the metal, a connection implied above. Lastly, we compare the predictions of the MD simulations with reported experimental observations.

2. Simulation model

A primary difficulty in establishing fundamental mechanisms of SPD from experimental studies alone is the inability to perform measurements that can systematically isolate different alloy properties. For example, experimental SPD studies performed on a series of Cu alloys, e.g., Cu-Ag, Cu-Co, Cu-Nb, etc., have provided control of the heat of mixing, but the elastic properties and crystalline structures of these systems also vary. Such issues can be largely avoided in MD simulations by the choice of interatomic potential. Specifically, in the present work we create model pseudo-alloys, such as Cu(1)-Cu(2), whereby the interactions between Cu(1) atoms are identical to those between Cu(2) atoms, with both represented by those of actual Cu. On the other hand, the interactions between Cu(1) and Cu(2) atoms are varied to change the heat of mixing, while maintaining the same elastic properties. The many-body Cleri-Rosato potentials, which are based on the second-moment approximation to the tight-binding scheme, have been shown to be well-suited for this purpose [29,30]. The system energy in this model has the form [31],

$$E_c = \sum_{i=1}^N \left\{ \sum_{j=1}^N A_{\alpha\beta} \exp \left[-p_{\alpha\beta} \left(-\frac{r_{ij}}{r_{\alpha\beta}^0} - 1 \right) \right] \right. \\ \left. - \sqrt{\sum_{j=1}^N \xi_{\alpha\beta}^2 \exp \left[-2q_{\alpha\beta} \left(-\frac{r_{ij}}{r_{\alpha\beta}^0} - 1 \right) \right]} \right\} \quad (4)$$

where α and β refer to the chemical identity of atoms i and j , and $r_{\alpha\beta}^0$ is nearest neighbor distance in the $\alpha\beta$ lattice. Parameter values for the metals of interest here are found in ref.[31] and reproduced in the supplementary material (see Table S1-A). Important here is that by varying the parameter $\xi_{\alpha\beta}$ in Eq. (4), the heat of mixing in a binary alloy can be systematically changed, while leaving the elastic properties of the alloy nearly the same. This is illustrated for dilute Cu(1)-Cu(2) pseudo-binary alloys in Fig. 1, where the elastic constants and the heat of mixing parameter, $\Omega = \Delta H_{mix}/X_{Cu(1)}X_{Cu(2)}$, are plotted as a function of $\xi_{\alpha\beta}$. The value of Ω corresponding to a given value of $\xi_{\alpha\beta}$ was more-or-less constant over the entire composition (see supplementary Fig. S1). We refer to the Cu(1)-rich and Cu(2)-rich phases as γ_1 and γ_2 , respectively. Similar pseudo-alloys were prepared using Cleri-Rosato interatomic potentials for other FCC metals to examine the role of the elastic constants.

We implement our model using the molecular dynamics (MD) code LAMMPS [32] to investigate the microstructural evolution of binary alloys during shear deformation. The deformation is applied as cyclical, bi-axial compressions along two prescribed orthogonal axes for a strain of $\sim 20\%$ along each axis and at a total strain rate of $\dot{\epsilon} = 5 \times 10^9$ s⁻¹. Varying the strain rate between 10^8 s⁻¹ and 5×10^9 s⁻¹ was found to have only minor effect on the results. The crystal was oriented with $\langle 100 \rangle$ directions aligned with the x, y, and z axes of the simulation cell, and the two axes along which the deformation is applied are switched after each compression step with an overall period of nine steps

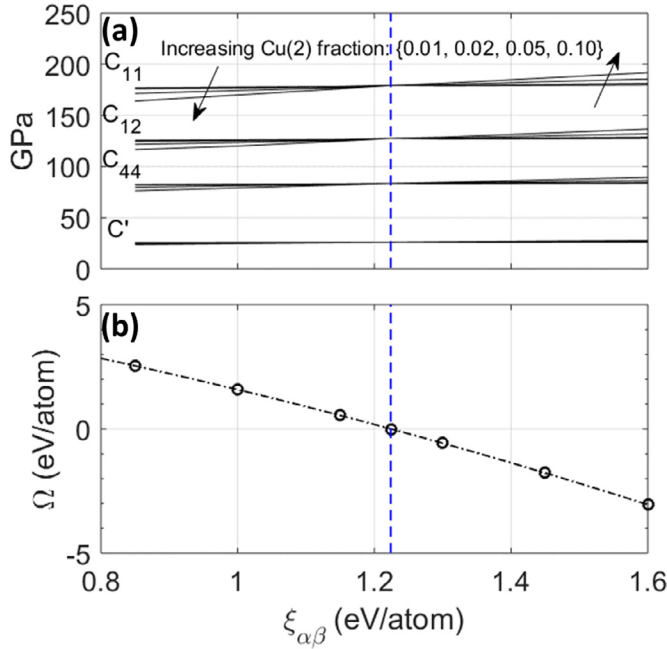


Fig. 1. Variation of (a) the elastic constants and (b) the heat of mixing, Ω (eV/atom) of the Cu(1)-Cu(2) binary system with the hopping integral parameter, $\xi_{\alpha\beta}$, defined by the Clari-Rosato potential. The dashed vertical line corresponds to pure Cu. Note that $C' = (C_{11} - C_{12})/2$ is the tetragonal shear modulus. The heat of mixing values are constant over the entire compositional range; whereas, increasing the nominal compositions, as indicated by the arrows in (a), had a small effect on the elastic constants. For example, for a very large heat of mixing of $\Omega = 2$ eV/atom, the resultant effect on the elastic constants was a decrease of $\sim 0.7\%$ per addition of 1 at.% Cu(2).

comprising an applied strain cycle. Considering the ‘free’ axis for which $\sigma_{ii} = 0$, the nine-step bi-axial compression cycle follows the path $\{y, x, z\}$, $\{x, z, y\}$, $\{z, y, x\}$ (see supplementary Fig. S0 for a schematic of this process). The exact strain along each axis is varied somewhat to maintain the same average dimensions of the

cell after every three compressive steps. We employ this complex deformation scheme to avoid shear localization. The effect of box size was examined by comparing the results from simulation cells with length along each dimension of $L = 28$, $L = 40$, or $L = 60$ lattice parameters (a_0), or in terms of the total number of atoms in the system, 87808, 256000 or 864000 atoms, respectively.

In Fig. 2 we illustrate the effect of varying $\xi_{\alpha\beta}$ (effectively Ω) on the steady-state alloy microstructures of equiatomic Cu(1)-Cu(2) and Pt(1)-Pt(2) alloys during shear deformation. Plotted in this figure are the short-range order (SRO) parameters after steady-state shearing for the two alloy systems. Here the expression for the SRO parameter is taken from ref. [22]

$$SRO = \frac{\left(\frac{c}{1-c}N_A^B + \frac{1-c}{c}N_B^A\right)}{N_A^A + N_B^B} - 1 \quad (5)$$

where N_A^B is the number of nearest neighbor B atoms around A atom (and vice versa for N_B^A), and c is the atomic fraction of A atoms in the A - B alloy. Also shown are the corresponding equilibrium values of the short-range order parameter, SRO_{eq} , at various temperatures obtained using canonical MC simulations, also implemented in LAMMPS [32]. The important features of Fig. 2(a) are that the SRO begins to drop rapidly in these alloys for values of $\Omega \geq 1.0$ eV/atom, and at a value of $\Omega \sim 2$ eV/atom, two-phase microstructures develop. This latter feature is observed visually in the snapshots of the atom positions presented in Fig. 2(b). Comparison of SRO following steady-state shearing with SRO_{eq} indicates an effective temperature of $\sim 7,000$ K when $\Omega < 1.0$ eV, but a lower temperature of $\sim 4,000$ K for Cu and ~ 5000 K for Pt when $\Omega \sim 2.0$ eV. The apparent drop in SRO and T_{eff} for $\Omega > 1.0$ eV correlates with the onset of γ_2 precipitation, but this will be discussed in detail below. As the present work is concerned primarily with the evolution of highly-immiscible systems characterized by a two-phase coexistence at steady state, we select $\Omega = 2$ eV/atom for Cu and Pt in the remainder of this work. Pseudo-binary alloys involving other FCC metals (Ag, Pd, Ni, and Rh) are also considered to explore the effect of the elastic properties on shear-induced chemical mixing.

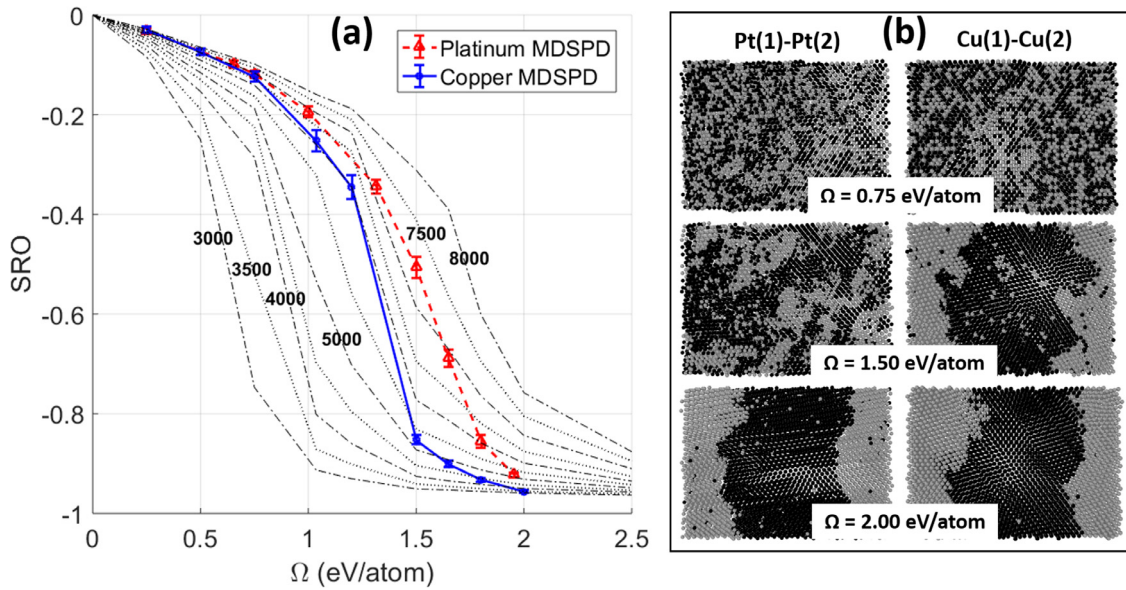


Fig. 2. (a) Short-range order (SRO) parameter as a function of the heat of mixing parameter, Ω (eV/atom), for equiatomic copper (solid blue line) and platinum (dashed red line) pseudo-alloys after steady-state shearing via the molecular dynamics severe plastic deformation (MDSPD) method detailed in the text. Also plotted are the equilibrium short-range order parameter values, SRO_{eq} (alternating dashed and dashed-dotted black lines), at temperatures ranging from 3000 K to 8000 K in increments of 500 K. (b) Shown are cross-sectional slices of the computational cell after a steady state has been attained for both Pt(1)-Pt(2) and Cu(1)-Cu(2) alloys at the indicated values of Ω (eV/atom).

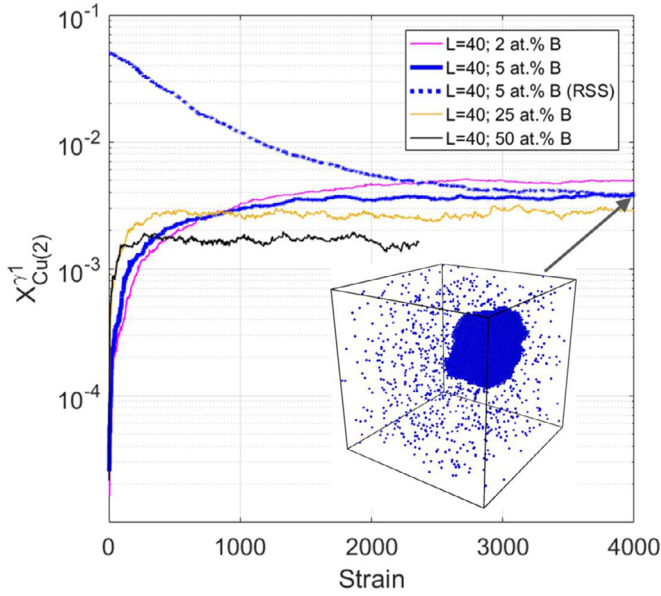


Fig. 3. Evolution with strain of the Cu(2) molar fraction in the γ_1 matrix, $X_{Cu(2)}^{\gamma_1}$, for a model alloy with varying nominal compositions. Most runs were initialized with a single spherical Cu(2) particle in a pure Cu(1) matrix, using a cell size with dimensions of 40 lattice parameters ($L=40$), corresponding to 256,000 atoms. For the 5 at.% alloy, the evolution starting from a random solid solution (RSS) is also shown (dashed blue line). Inset shows the final microstructure for the 5 at.% Cu(2) system featuring a single precipitate in equilibrium with a background steady-state solubility.

3. Results

3.1. Steady-state microstructures during low-temperature SPD

3.1.1. Dependence of the steady-state microstructure on the initial state of the system:

We begin our description of the shear-induced chemical mixing behavior of the Cu(1)-Cu(2) alloy with $\Omega = 2$ eV/atom by illustrating that the steady-state microstructure is independent of the initial state of the alloy. The results are reported in Fig. 3 for a Cu(1)-5 at.% Cu(2) alloy (bold, blue lines) where the molar fraction of Cu(2) in the γ_1 matrix, $X_{Cu(2)}^{\gamma_1}$, is plotted as a function of strain for two initial conditions, either a random alloy or a single Cu(2) precipitate in the pure γ_1 matrix. This figure shows that following low temperature shearing, the steady-state molar fraction converges to the same value which is referred to here as the *steady-state solubility*, $X_{Cu(2)}^{\gamma_1,SS} = 0.38$ at.% Cu(2). The steady-state microstructure in each case, moreover, contains just a single γ_2 precipitate with radius $r \approx 3.25$ nm. The inset in Fig. 3 illustrates that the precipitate morphology is compact and very nearly equiaxed. We note that for a given nominal alloy composition, the precipitate size, steady-state solubility, and total strain to reach this steady state are dependent on the size of the computational cell, but these details are discussed below. Also shown in Fig. 3 are $X_{Cu(2)}^{\gamma_1}$ versus strain curves for alloys with other global concentrations of Cu(2), each converging to their corresponding steady-state solubility, $X_{Cu(2)}^{\gamma_1,SS}$. Thus, the steady-state solubility is observed to *decrease* systematically with *increasing* average concentrations of Cu(2). Moreover, only a single, compact precipitate is present in each alloy when the system reaches steady state, similar to the 5 at.% Cu(2) alloy.

3.1.2. Dependence of steady-state solubilities on precipitate size: Gibbs-Thomson-like behavior during low-temperature SPD

Since only one precipitate remains within the computational cell at steady state, the relationship between steady-state solubil-

ity and precipitate size is directly available from data like those plotted in Fig. 3. These results are collected in Fig. 4(a) where the steady-state solubilities in both the matrix, $X_{Cu(2)}^{\gamma_1,SS}$, and within precipitate, $X_{Cu(1)}^{\gamma_2,SS}$, are plotted as a function of inverse precipitate radius. We emphasize that this set of results is *not* dependent on the size of the computational cell. The alloy concentration at $r^{-1} = 0$ was determined using an equiatomic alloy and initiating the cell with a planar interface. In this case the interface remains flat in steady state, albeit with some roughness (see Fig. 2(b)). Notably, initiating the computational cell of this equiatomic alloy as a random solution leads to the same final microstructure, again with a flat interface, which is consistent with our conclusion that the final microstructure is independent of the starting conditions. In this latter case, spherical precipitates first nucleate, but as they grow larger and agglomerate, the high $\gamma_1 - \gamma_2$ interface energy, in combination with the periodic boundary conditions and cell size, favors a flat interface. The significant feature in Fig. 4(a) is that the concentration of Cu(2) is an increasing function of r^{-1} , in both the γ_1 and γ_2 phases, as expected for two-phase alloys in thermal equilibrium when capillary effects are significant [33], i.e., the shearing behavior is reminiscent of the Gibbs-Thomson (G-T) effect, which for dilute solutions can be expressed,

$$X(r) = X(r = \infty) \cdot \exp\left(\frac{2\sigma V_0}{rRT}\right) \quad (6)$$

where σ represents the $\gamma_1 - \gamma_2$ interfacial free energy and V_0 is the atomic volume. We further illustrate such G-T like behavior in Fig. 4(b) where the evolution of a system initiated with one large γ_2 precipitate ($r = 2.5$ nm) and four smaller ones ($r = 1.1$ nm), embedded in a γ_1 matrix is followed. The simulation is initialized with $X_{Cu(2)}^{\gamma_1} = 0.45$ at.% Cu(2), which is the steady-state solubility of the larger precipitate, but undersaturated for the smaller ones. As Fig. 4(b) reveals, the larger precipitate grows at the expense of the 4 smaller ones until only a single precipitate remains, and the steady-state solubility is slightly reduced.

The data in Fig. 4(a) can be used to determine an effective temperature during SPD in two ways. The more direct method uses the alloy solubility for this purpose. For the dilute solutions involved here, we use the regular solution model and determine T_{eff} from the solubility data for the case $r^{-1} = 0$, i.e.,

$$X_{Cu(2)}^{\gamma_1,SS}(r = \infty) = \exp\left(-\frac{\Omega}{RT}\right). \quad (7)$$

We note canonical MC simulations support use of the regular solution at the low concentrations observed in Fig. 4 (see supplementary information, Fig. S1). From Fig. 4, $X_{Cu(2)}^{\gamma_1,SS}(r = \infty) \approx 0.002$ and with $\Omega = 2$ eV/atom, Eq. (7) yields $T_{eff} \approx 3700$ K. This value agrees well with the effective temperature deduced for Cu using the short-range order parameter, SRO, when $\Omega = 2.0$ eV/atom, as reported in Fig. 2. We can also determine the effective temperature using Eq. (6), assuming the G-T relation is valid during SPD, although Eq. (6) requires first determining the interface energy, σ .

We determined σ using two methods. The first method employs canonical MC simulations to determine $X_{Cu(2)}^{\gamma_1,SS}$ as a function of precipitate size, and then uses the G-T relation to extract a value for σ at a particular temperature. The second method approximates the interfacial free energy by the enthalpy term and calculates this enthalpy directly for the precipitates in the sheared systems. Both methods yield a value of $\sigma \approx 1.85 \pm 0.05$ J/m² (see supplementary information, Fig. S2). The use of this value in Eq. (6) yields the effective temperatures, $T_{eff} = 1400 \pm 25$ K (using the steady-state solubilities in the γ_1 matrix, $X_{Cu(2)}^{\gamma_1,SS}(r)$) and 2300 ± 200 K (using the steady-state solubilities within the γ_2 precipitate, $X_{Cu(1)}^{\gamma_2,SS}(r)$), as seen in Fig 4(a). While these values are an

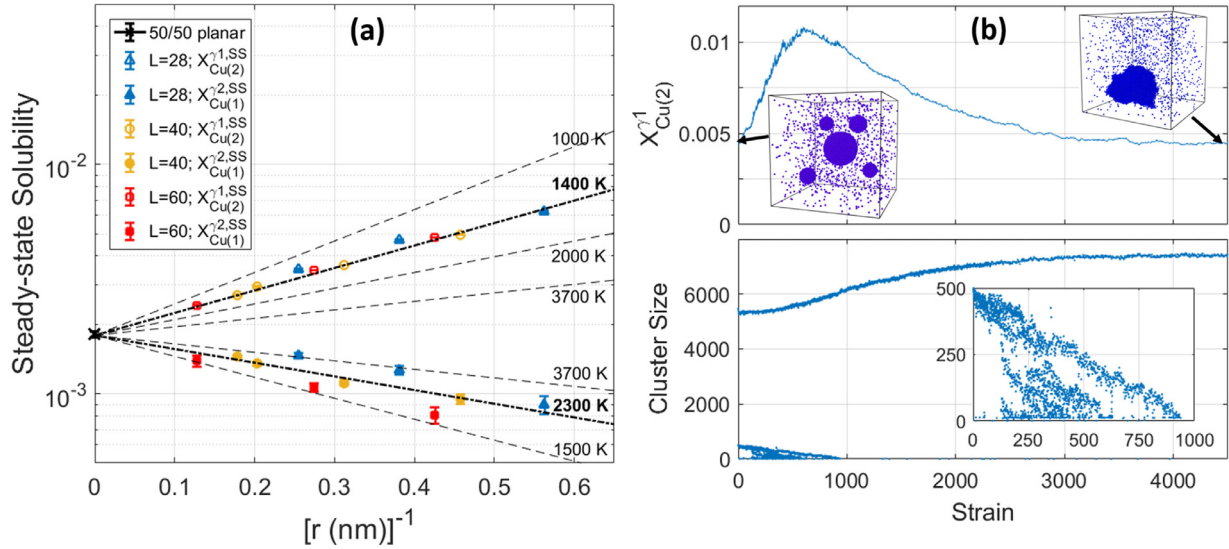


Fig. 4. (a) Steady-state solubilities in the matrix (open symbols), or within the precipitate (closed symbols), as a function of the reciprocal precipitate size, r^{-1} (nm^{-1}). Data is shown for the three different simulation cell sizes investigated in this work, where L is the number of lattice planes along each dimension. The dotted lines are expected steady-state solubilities from the Gibbs-Thomson relation (see Eq. (6)) at the indicated temperatures using a value for the interface free energy of $\sigma = 1.85 \text{ J/m}^2$, which was itself determined using two different approaches (see text for details). (b) Illustration of the Gibbs-Thomson effect during shearing for a system initialized with one precipitate of size $r = 2.5 \text{ nm}$ and four smaller ones ($r = 1.1 \text{ nm}$), and an initial background molar fraction of $X_{\text{Cu}(2)}^{\gamma_1} \approx 0.45 \text{ at.}\%$ Cu(2) corresponding to the steady-state solubility of the larger precipitate alone. Top panel shows the evolution with strain of the background molar fraction of Cu(2); whereas, the lower panel shows the sizes of all clusters.

order of magnitude greater than the actual ambient temperature of 100 K, they are still lower than the temperatures deduced from the solubility data and the short-range order ($T_{\text{eff}} \sim 3,700 \text{ K}$); moreover, the two values are also somewhat different from each other.

3.2. Kinetics of phase evolution during SPD

3.2.1. Precipitate dissolution during SPD

Previous MD simulations have explored how various nanoprecipitates dissolve in Cu during SPD, (Ag, Ni, Fe, Nb, V) [34]. For Ag and Ni, the heat of mixing with Cu is small (compared to $\Omega = 2 \text{ eV/atom}$) and the precipitates are coherent. In these systems, dislocations transfer across the precipitate/matrix interface, repeatedly cutting the precipitate, and eventually dissolving the precipitate. Fe is BCC, but after a short transient period, the small Fe precipitate transforms to FCC and behaves similarly to Ag and Ni. The BCC Nb and V precipitates have very high heats of solution in Cu, $\Omega > 1 \text{ eV/atom}$, and they are not coherent in Cu; dislocations do not transfer across the interface. For the small precipitates that could be investigated by MD, $r < 8 \text{ nm}$, little dislocation activity was observed in either the V or Nb nanoparticles. In these systems, dissolution occurred by an amorphization process at the interface followed by chemical mixing. From this interface region, V and Nb atoms entered the Cu matrix and homogenized, but Cu did not dissolve into the V or Nb precipitates, presumably, owing to lack of dislocation activity in the precipitates. For the present Cu(1)-Cu(2) system with $\Omega = 2 \text{ eV/atom}$, the Cu(2) precipitate has a high heat of mixing with the matrix, but it does remain coherent with the matrix, and the elastic properties of the precipitate and matrix are the same. The dissolution behavior in this system, accordingly, is intermediate between the two extremes of FCC and BCC precipitates described above; i.e., dislocations do not cross the $\gamma_1 - \gamma_2$ interface, but they are active in both phases. This behavior is illustrated by the MD snapshots presented in Fig. 5. In Fig. 5(a-d) the glide paths of partial dislocations are shown for Cu(1)-Cu(2) systems with heats of mixing of 1 eV (Fig. 5(a,b)) and 2 eV (Fig. 5(c,d)), and each system contains a single precipitate with a 3.3 nm diameter. These glide paths, which are revealed by identifying atoms

with HCP symmetry (see refs. [29,35] for details), are observed to transfer across the precipitate/matrix interface when $\Omega = 1 \text{ eV}$ (Fig. 5(a)), but not when $\Omega = 2 \text{ eV}$ (Fig. 5(c)). Also shown in these figures are atoms with very low symmetry (colored black). For the precipitate with $\Omega = 2 \text{ eV/atom}$, the interface has many such low symmetry atoms ($\sim 40\%$ of interface atoms are on disordered sites in Fig. 5(c) versus $\sim 20\%$ for the 1 eV system in Fig. 5(a)), illustrating that a high degree of structural disorder at the interface facilitates local chemical mixing during shear as described for Cu/BCC precipitate interfaces in refs [34,36]. Unlike the BCC precipitates in Cu, however, the disordered interface does not broaden, the composition profile remains sharp, and dissolution of Cu(1) in the γ_2 precipitate is nearly the same as that for Cu(2) alloy into the γ_1 matrix, as noted in section 3.1.2, above. Lastly, Figs. 5(e-h) show that even for the alloy with $\Omega = 2 \text{ eV}$, dislocations begin transferring across the $\gamma_1 - \gamma_2$ interface when the precipitate radius falls below 2 nm, cutting the precipitate, as should be expected [37].

The initial dissolution rate of pure Cu(2) precipitates in a pure Cu(1) matrix, $(dN_{\text{Cu}(2)}^{\gamma_1}/d\epsilon)_{\epsilon=0}$, was determined as a function of precipitate size by plotting $N_{\text{Cu}(2)}^{\gamma_1}$, the number of Cu(2) atoms entering the γ_1 phase, as a function of strain and measuring the slope at zero strain (see Fig. S3 in the supplementary material for details). These results are shown inset in Fig. 6. For such an undersaturated system, the initial dissolution rate for systems at elevated temperature is expected to scale with the interface area and the driving force for dissolution [38],

$$k = 2(X_{\text{Cu}(2)}^{\gamma_1}(r) - X_{\text{Cu}(2)}^{\gamma_1})/(X_{\text{Cu}(2)}^{\gamma_2} - X_{\text{Cu}(2)}^{\gamma_1}(r)), \quad (8)$$

where $X_{\text{Cu}(2)}^{\gamma_1}$ is the molar fraction of Cu(2) in γ_1 , adjacent to the interface between the matrix and the precipitate of radius r ; $X_{\text{Cu}(2)}^{\gamma_2}$ is taken from Fig. 4 above, i.e., $X_{\text{Cu}(2)}^{\gamma_2}(r) = X_{\text{Cu}(2)}^{\gamma_1, \text{SS}}(r)$. Initially, the γ_1 matrix consists of only Cu(1) atoms, so $X_{\text{Cu}(2)}^{\gamma_1} = 0$. Furthermore, the γ_2 precipitate is initially pure Cu(2), so $X_{\text{Cu}(2)}^{\gamma_2} - X_{\text{Cu}(2)}^{\gamma_1}(r) \approx 1$ as $X_{\text{Cu}(2)}^{\gamma_1}(r) \ll 1$. Thus, the above expression for the dissolution driving force simplifies to $k \approx 2X_{\text{Cu}(2)}^{\gamma_1, \text{SS}}(r)$. From the main plot in Fig. 6, it is seen that the data follow the expected scaling quite well for

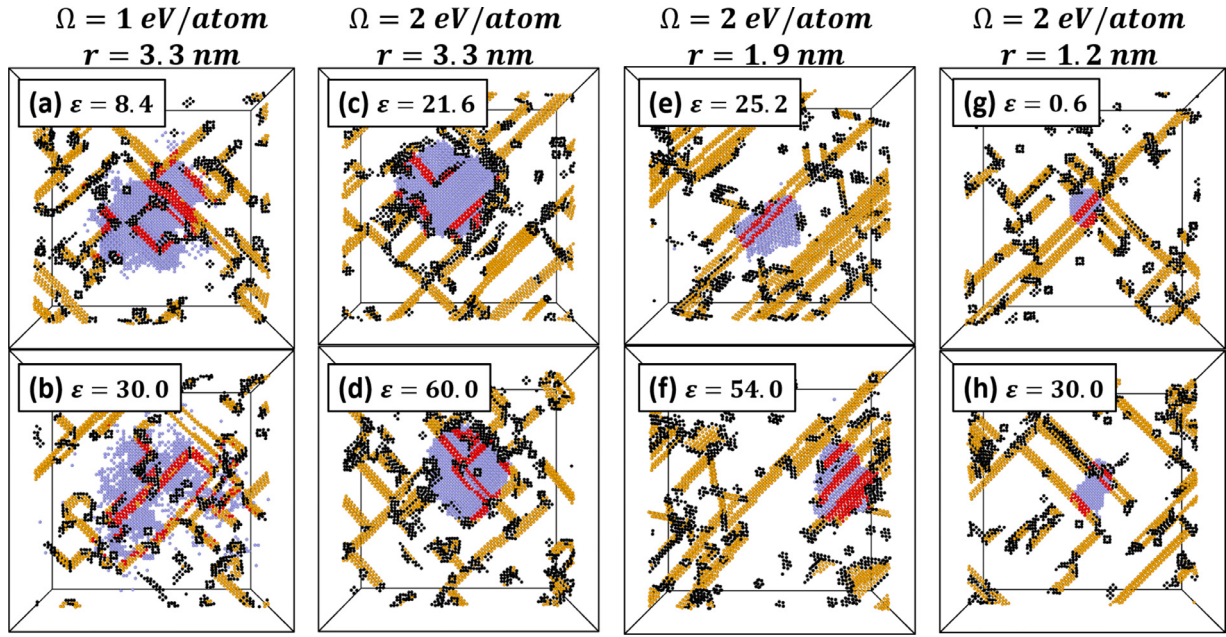


Fig. 5. Shown are 0.5 nm width slices of the simulation cell for alloys with a heat of mixing of either (a)-(b) $\Omega = 1$ eV/atom or (c)-(h) $\Omega = 2$ eV/atom after the indicated amount of applied strain, ϵ , using the MD deformation scheme. The colorization corresponds to the local symmetry around each atom according to the common neighbor analysis (CNA) method implemented in OVITO [35]. FCC lattice sites with Cu(1) atoms have been omitted for clarity; whereas, those with Cu(2) atoms are light blue. Low symmetry, disordered sites are colored black. HCP sites with Cu(1) atoms (yellow) or Cu(2) (red) correspond to the glide paths of partial dislocations. All systems were initialized with a single γ_2 precipitate with the indicated radii in a γ_1 matrix.

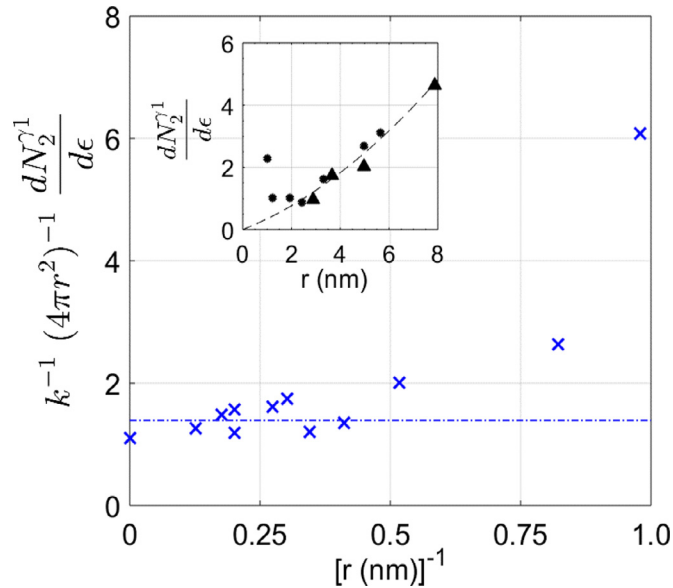


Fig. 6. Erosion rates, $\frac{dN_2^{21}}{d\epsilon}$, of Cu(2) atoms from the γ_2 precipitate during shear deformation normalized by the precipitate surface area and the driving force for dissolution, k (see text for details). These rates were determined for an initially phase-separated system, i.e. the γ_1 matrix is initially depleted of any solute atoms and N_2^{21} is the number of Cu(2) atoms in the γ_1 phase. Note that the value at $r^{-1} = 0$ was calculated from the erosion rate of Cu(2) atoms from a planar interface, $(\frac{dN_2^{21}}{d\epsilon})_{r=0} = 4 \times 10^{-3} \text{ nm}^{-2}$, which was constant for the three different cell sizes considered in this work. The horizontal dashed-dotted line is a guide for the eye to distinguish between the two regimes observed for the erosion rates where precipitates of size $r < 2 \text{ nm}$ ($r^{-1} > 0.5 \text{ nm}^{-1}$) show an enhanced erosion rate. Finally, plotted in the inset is the actual erosion rate as a function of the precipitate size, and results are shown for two simulation cell sizes, $L=40$ (circles) or $L=60$ (triangles).

$r > 2 \text{ nm}$, but for $r < 2 \text{ nm}$, the normalized erosion rates increase rapidly with decreasing radius. From the description of Fig. 5, it is clear that this observed enhanced dissolution rate at small r derives from dislocations shearing the small precipitates.

3.2.2. Nucleation and growth of precipitates during low temperature SPD

The nucleation of γ_2 precipitates in initially random Cu(1) alloys was examined by following the size of individual Cu(2)-rich clusters as a function of strain. Fig. 7(a), for example, plots the size of the largest cluster versus strain. For the alloys with a global composition $> 3 \text{ at.}\%$ Cu(2), the size of the largest cluster increases more or less steadily with strain from the outset, whereas for alloys with concentrations $\leq 3 \text{ at.}\%$, the size quickly plateaus at quite small sizes. Also notable in this figure are the large fluctuations in the size of these clusters, especially in the more dilute alloys. The rapid changes in cluster size correspond to the clusters being fragmented, for example by a dislocation shuffling mechanism [39,40], or through clusters agglomerating into a larger cluster as illustrated in Fig. S4 of the supplementary information.

To better understand why the 3 at.% alloy fails to nucleate, we estimate the *effective critical size* for nucleation in this alloy by embedding 10 clusters of Cu(2) atoms of size N within a random 3 at.% Cu(2) alloy before activating shear deformation. By effective critical size we imply only that at least one of the 10 embedded clusters will nucleate a precipitate on the time and length scale of these simulations. The results are illustrated in Fig. 7(b). Only clusters with more than ~ 80 Cu(2) atoms ($r \geq 0.6 \text{ nm}$) are found to grow. As seen in Fig. 7(a), none of the clusters in the initially random 3 at.% Cu(2) alloy reach this critical size for nucleation. As the alloy concentration increases, the critical radius decreases. For example, in the 5 at.% Cu(2) alloy, clusters of ~ 6 –7 Cu(2) atoms were observed to grow continuously by absorbing solutes or agglomerating with other small clusters. It is worth noting that classical nucleation theory (CNT), using a regular solution model [41] predicts a critical radius for the 5 at.% alloy of just 0.29 nm (~ 10 atoms) at 3,700 K, suggesting that there is only a limited energy barrier to nucleation, if any, at these larger alloy concentrations.

The results for precipitate growth are presented in Fig. 8. In Fig. 8(a) we plot the number of clusters/precipitates present in the γ_1 matrix as a function of strain for alloys that were initiated as random solutions; the concentration of Cu(2) ranges from

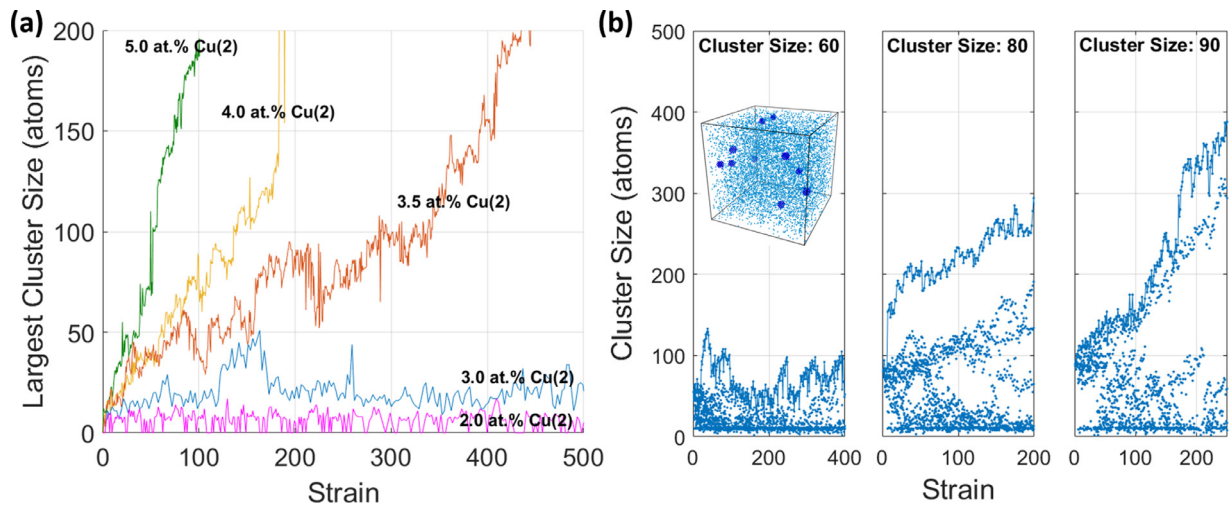


Fig. 7. (a) Size of the largest cluster as a function of strain for Cu(1)-Cu(2) alloys with the indicated nominal compositions and initialized as a random solid solution. (b) Cluster size evolution with strain of a 3 at.% Cu(2) random solid solution seeded with clusters of either 60 atoms, 80 atoms or 90 atoms, corresponding to the panels from left to right. Shown in the first panel inset is the corresponding initial configuration. Note that on the second panel, there is a large jump in the size of one cluster at strain ~ 10 due to two seed clusters agglomerating.

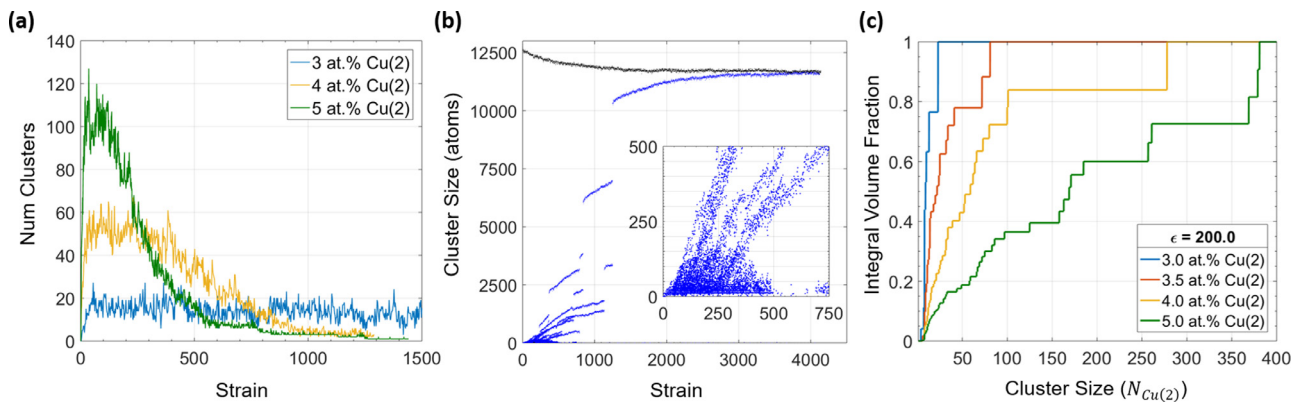


Fig. 8. (a) Total number of clusters with increasing strain for simulation cell initialized as a random solid solution with the indicated nominal compositions. (b) For the 5 at.% Cu(2) alloy initialized as either a solid solution (blue) or a single precipitate (black), the evolution of all cluster sizes are plotted as a function of strain; Inset shows the cluster sizes at early strain values with greater resolution. (c) Integral volume fraction of Cu(2) atoms in clusters up to size $N_{Cu(2)}$ as a function of $N_{Cu(2)}$ after strain of 200 for systems with the indicated nominal compositions and initialized as a random solid solution.

3 at.% to 5 at.%. For this figure, Cu(2) atoms were considered as part of a cluster when 50% of their nearest neighbors were either Cu(2) atoms or were themselves ‘bonded’ to another Cu(2) atom that met this criterion. Thus, the minimum size for a cluster was 6–7 Cu(2) atoms. For the 4 and 5 at.% alloys, the number of clusters increases rapidly with strain, reaching a maximum after strains of 100 and 50, respectively, and then rapidly decreases. For the 3 at.% Cu(2) system the number of clusters initially increases as well, but it quickly plateaus, which is consistent with the lack of nucleation of precipitates, even after extended values of strain (>1500). In Fig. 8(b) the sizes of individual clusters are tracked as a function of strain for the 5 at.% Cu(2) alloy, starting from a random solution (blue line). Two behaviors are clearly observed, the first being a more-or-less continuous growth of cluster size and the second showing large discrete jumps in size; the latter mechanism becomes more dominant as the cluster size distribution shifts to larger sizes (at higher values of strain). These growth mechanisms are attributed to the convective atomic transport mechanism that underpins severe plastic deformation. As described elsewhere [29,42], atoms move relative to one another by dislocations passing between them. Consequently, the relative motion of clusters of atoms (including single atoms) is nearly independent of their size, and single atoms can join clusters, or clusters can agglomerate. At

low values of strain, most Cu(2) atoms are monomers or comprise small clusters, so the growth of clusters appears continuous. At larger values of strain, many of the Cu(2) atoms are found in clusters, thus cluster agglomeration begins to dominate. For example, in Fig. 8(c), the integral volume fraction of Cu(2) atoms contained in precipitates up to size $N_{Cu(2)}$ is plotted as a function of $N_{Cu(2)}$. At a strain of 200, the distribution is roughly linear, indicating that Cu(2) atoms are distributed equally in clusters of all sizes (by volume). Since the relative motion of clusters is nearly independent of cluster size, and the cross section for cluster-cluster collisions varies only weakly with size ($\sim N^{2/3}$), clusters of all sizes contribute to cluster growth almost equally, explaining both the continuous and discrete jumps in precipitate growth.

3.3. Dependence of the effective temperature on the shear modulus

In the introductory section we suggested that the effective temperature could be estimated by considering the work done on an atom during plastic deformation, i.e., $\sigma_{th} \cdot \epsilon \cdot V_0$. Since the theoretical strength is related to the shear modulus, we should expect the effective temperature to be related to the elastic constants. We tested this hypothesis by repeating the simulations performed on Cu(1)-Cu(2) on other FCC metals (Ag, Cu, Ni, Pd, Pt and Rh) for

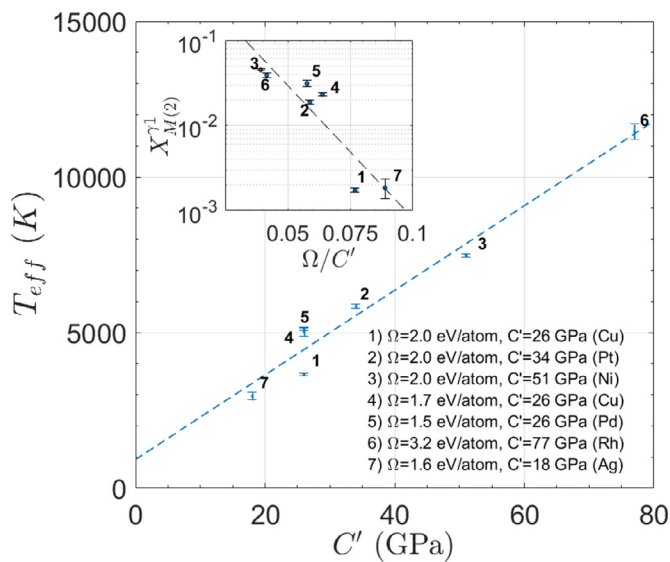


Fig. 9. Analogous to the Cu(1)-Cu(2) system, the effective temperature during steady-state shear deformation is determined for various equiatomic immiscible M(1)-M(2) alloys, where M={Ag, Cu, Ni, Pd, Pt, Rh}, and shown to scale with the tetragonal shear modulus, C' (GPa), of the matrix. Inset shows the actual steady-state solubilities, $X_{M(2)}^{Y_1}$, achieved for a given choice of system parameters, i.e. the heat of mixing, Ω (eV/atom), and the FCC metal, M, with its corresponding elastic properties.

which Cleri-Rosato potentials are also available (see ref. [31] or the supplementary Table S1-A) to determine the steady-state solubility. For the more refractory materials, such as Rh, the effective temperature is so high that even with $\Omega = 2$ eV/atom, Rh(1)-Rh(2) alloys become single-phase solutions during SPD. In Ag, T_{eff} is very low and the steady-state solubility is too low to determine. We thus have obtained a more comprehensive curve for FCC metals by using different values of Ω , as shown in Fig. 9. Here, using the expression for solubility in the regular solution model, Eq. (7), we plot T_{eff} vs. C' , where $T_{eff} = \Omega / k \ln(X_{M(2)}^{Y_1})$, and $X_{M(2)}^{Y_1}$ is the steady-state solubility corresponding to the particular choice of the heat of mixing parameter, Ω (eV/atom), and the FCC metal, M, with its corresponding value for the tetragonal shear modulus, C' (GPa). Overall, the fit in Fig. 9 is quite satisfactory considering the complexity of plastic deformation.

4. Discussion

The results of this MD study provide a detailed description of the atomistic processes involved in phase evolution during low-temperature SPD, and they clarify the significance of the process temperature, T_0 in the (modified) effective temperature model. Overall, the description of T_0 as a process temperature successfully captures many of the main features in low-temperature SPD. It rationalizes the origin of the two-phase microstructure during low temperature SPD in strongly immiscible alloys, Gibbs-Thomson-like (G-T) behavior and precipitate coarsening, and why the steady-state microstructures are independent of the initial state of the alloy. The correlation between the magnitude of T_0 and the elastic properties of the alloy, moreover, provides a physical basis for the model. It is also worth commenting on the general agreement between the findings of the present MD simulations and SPD experiments on highly immiscible Cu-based alloys. Recent high pressure torsion (HPT) experiments performed at low temperatures on dilute Cu-Nb and Cu-Mo alloys report steady-state solubilities of ~ 2 at.% and 1 at.% for Nb and Mo, respectively, yielding an effective temperature of $\sim 3,000$ K for these systems [6,8]. This value is in good agreement with the value obtained by MD simulations on

Cu(1)-Cu(2), 3,700 K, particularly in light of the differences in the alloy systems, strain rates, and strain paths employed in the simulations and experiments. The HPT studies on these Cu-Nb and Cu-Mo alloys also showed a characteristic strain of $\sim 1,000$ for solute to precipitate out of solution, which agrees quite well with that shown in Fig. 3 for a 5 at.% Cu(1)-Cu(2) alloy. The good agreement is possibly somewhat fortuitous considering again the difference in systems and driving forces; nevertheless, it offers a reassuring confirmation that the simulations capture the underlying physical mechanisms of SPD.

Despite the general agreement between the METM with simulations and between the simulations and experiments, a few of the above observations require comment: (i) the values of T_0 deduced from the SRO parameter at low heats of mixing, $\Omega < 1.4$ eV, are greater than those for $\Omega > 1.4$ eV; (ii) T_0 deduced from the G-T relation appears too low both in the matrix and in the precipitate; (iii) precipitates do not appear to nucleate in the ≤ 3 at.% alloys, at least not on the expected time scales when compared to the 4 at.% and 5 at.% alloys using classical nucleation theory; (iv) experiments reveal that both Cu-Nb and Cu-Mo undergo self-organized microstructural development, i.e., the steady-state microstructure featured size-selected nanoprecipitates of either Nb or Mo, approximately 15 nm in diameter [6,8]. This microstructure, moreover, was observed independent of the conditions of the alloy prior to SPD, i.e., large precipitates smaller than the characteristic patterning length became larger, whereas larger precipitates became smaller, thus the system does not continue to coarsen. While these experimental observations of self-organization are not directly in disagreement with the simulations, as the computational cell for MD is smaller than 15 nm, they are, nevertheless, in disagreement more generally with the METM, which predicts coarsening.

We begin discussion of these apparent discrepancies between the METM and the simulation results by first considering the deformation process. Previous MD simulations have shown that shear-induced chemical mixing in crystalline metals is mediated by convective diffusional flows, such as dislocation glide [29], and as long as dislocations glide homogeneously along slip planes, atoms move randomly. Heterogeneities in the microstructure, however, can alter this picture [34,43]. As shown in Figs. 6 and 7, dislocations can cut clusters of solute atoms and small precipitates, but not large ones, with the transition occurring at $r \approx 2$ nm for our Cu(1)-Cu(2) alloys. Consequently, the mechanism for cluster dissolution changes rapidly at this critical cluster size, thus explaining the increase in the dissociation rate reported in Fig. 7 for $r < 2$ nm. This shift in dissolution mechanism in small precipitates can clearly influence both SRO and nucleation. Fig. 2, for example, suggests that for alloys with low heats of mixing, which do not form precipitates during low temperature SPD, the SRO behaves as if the temperature were very high, $\sim 7,000$ K, as mixing tends to be random. Once the heat of mixing is sufficient to induce precipitation, however, dislocation motion is no longer random, rather it is guided by the interface, and the effective temperature drops to $\sim 3,700$ K. Since SRO is determined by the fraction of Cu(1)-Cu(2) bonds, and these bonds are mostly observed at precipitate-matrix interfaces, this deduced effective temperature refers to the interface region. We attribute the failure of the 1–3 at.% Cu(2) alloys to undergo precipitation to derive from this same underlying mechanism. If the alloy system is initiated from a random solution, i.e., absent any interfaces, it initially behaves as if it were at a higher temperature, possibly as high as $\sim 7,000$ K, although in the supplementary information (Fig. S5), we deduce an effective temperature of $\sim 5,000$ K for the 3 at.% Cu(2) system initialized as a random solid solution with a heat of mixing of $\Omega = 2$ eV/atom. At 5,000 K the solubility limit of the Cu(1)-Cu(2) alloys is ~ 1 at.%, and at 7,000 K it is even > 3 at.%. In contrast, when the system is already two-phase at the onset of shearing, the effective tem-

perature is 3,700 K and the solubility is $\sim 0.2 \text{ at.}\% \text{ Cu(2)}$ as shown in Fig. 4. In other words, the effective temperature in the random alloy is greater than that in the two-phase alloy, and this higher temperature reduces the supersaturation, slowing nucleation. Interestingly, classical nucleation theory predicts a critical size of ~ 100 atoms for this alloy at 5,000 K, which is quite close to the 80 atoms reported in Fig. 4(b).

It is perhaps not surprising that the creation of interfaces changes the local value of T_0 since, as shown above, the deformation mode at the interface is very different than that in the grain interior. We can only speculate why interfaces lower the effective temperature. For example, Fig. 6 shows that the matrix/precipitate interface is a source of dislocations, possibly owing to the high crystalline disorder on the interface or simply due to the high interface energy, i.e., weak bonding. This leads to a weak interface. Indeed, for a Cu(1)/Cu(2) bilayer sample with the normal to the heterophase interface oriented along the $[111]$ direction, applying simple shear along the $(111)/\langle 110 \rangle$ slip system resulted in yielding at the interface with the shear strength decreasing as the heat of mixing is increased (data not shown). Since the local effective temperature depends on the local stresses required for plastic deformation, we expect a lower value of T_0 at the interface.

While the above arguments can rationalize why precipitates in the initially random alloys with nominal compositions $\leq 3 \text{ at.}\% \text{ Cu(2)}$ were not observed to nucleate precipitates, it is worth considering this observation in more detail since it impacts directly on our conclusion that the steady-state microstructure is independent of the initial conditions. Cordero and Schuh, for example, concluded from a combination of Monte Carlo simulations and experimental observations that if the strengths of the two phases were sufficiently different, initially random solutions would not undergo precipitation, while phase-separated systems would remain two-phase [44]. A similar effect of the difference in phase strength on chemical mixing rates in two-phase systems during SPD was reported by Delogu using molecular dynamics simulations [30]. While we do not necessarily disagree with these conclusions, we attribute this outcome within the METM to the ballistic diffusion coefficient at the process temperature, $D_b^{T_0}$ in Eq. (2), simply tending toward zero in the hard phase so that steady state may not be obtained. The lack of dissolution of Al-W powders during ball milling [45] is a particularly illustrative example of a system with dissimilar strengths. A large difference in phase strength does not occur in the Cu(1)-Cu(2) system, and so this explanation does not explain the lack of nucleation here. We suspect, however, the reason again reflects the difficulty in reaching steady state. As noted above, the random 3 at.% alloys is presumably at an effective temperature on the order of 5,000 K initially, but γ_2 clusters containing as many as 50 Cu(2) atoms nevertheless form as indicated by Fig 7(a). It is therefore reasonable that on longer time/length scales, fluctuations will result in the formation of clusters larger than ~ 80 Cu(2) atoms, which is the effective critical size for nucleation observed in Fig. 7(b) for this background supersaturation. HPT experiments on Cu-Nb appear to agree with this hypothesis as initially random 2.5 at.% Nb alloys undergo precipitation during low-temperature SPD before coming to a steady-state Nb solubility of 1 at.%, indicating that nucleation can indeed occur in systems that are only moderately supersaturated during SPD [6].

The relatively low effective temperatures that are derived from the Gibbs-Thomson equation, 1400 K using $X_{\text{Cu(2)}}^{\gamma_1,SS}(r)$ and 2300 K using $X_{\text{Cu(1)}}^{\gamma_2,SS}(r)$ (see Fig. 4(a)) is more difficult to reconcile, but we point out that the Gibbs-Thomson effect is a quite subtle effect. Note that the effective temperature deduced from the Gibbs-Thomson relation in Fig. 4(a) is already ~ 14 times larger than the ambient temperature of 100 K, suggesting that a discrepancy of a factor of two in the effective temperature is relatively small. While

the statistical uncertainties in the calculated values of $X_{\text{Cu(2)}}^{\gamma_1,SS}(r)$ and $X_{\text{Cu(1)}}^{\gamma_2,SS}(r)$ are not sufficiently large to explain this discrepancy, systematic effects may significantly influence the results. For example, the normalized dissolution rates observed in Fig. 5 increase at small precipitate sizes due to dislocations beginning to cut the precipitates. This mechanism would tend to increase the matrix solubility as the precipitate size became smaller, thereby lowering the apparent effective temperature. This possibility is discussed in the supplementary material (see Fig. S6).

Lastly, we discuss briefly the observation that HPT processing of dilute Cu-Nb and Cu-Mo alloys leads to self-organized structures in steady state with precipitate sizes of $\sim 15 \text{ nm}$ [6,8]. This so-called “compositional patterning” appears to be in contradiction with the METM. While a detailed discussion of compositional patterning goes beyond the scope of this work, we remind the reader that precipitate growth derives from the convective transport of solute atoms, clusters and precipitates and their subsequent agglomeration. This transport mechanism, however, is limited by the grain size (see ref [43] for details), resulting in the growth mechanism becoming less efficient relative to precipitate dissolution when the precipitate size becomes comparable to the grain size. During HPT, the grain size in Cu-Nb and Cu-Mo is $\sim 50 \text{ nm}$ [6,8], while the precipitate size is $\sim 20 \text{ nm}$. As neither the original effective temperature model, nor the METM discussed here, takes into account the dependencies of the competing growth and dissolution dynamics on length scale, they are unable to predict compositional patterning.

5. Conclusions

In this paper we provided a detailed picture of phase evolution in strongly immiscible alloy during severe plastic deformation and evaluated how well modified effective temperature models capture this behavior. Many features of phase evolution revealed by the MD simulations can indeed be understood by the METM on a semiquantitative basis. The main successes are that the model explains phase separation and the surprising observation of Gibbs-Thomson-like behavior during SPD. The model is also consistent with the observation, at least in alloys with nominal compositions above 3 at.% Cu(2), that the steady-state microstructure is independent of the initial state of the alloy. Moreover, the actual magnitude of the effective temperature during low temperature shearing, T_0 , is roughly consistent with the work done by the shearing forces, and its scaling with the shear modulus in several metals is a notable success. The model, however, has some weaknesses as well. Most notably is that the exact value of T_0 depends on the local microstructure and shear mechanism, so that different regions of the microstructure evolve at different values of T_0 . The ETM of Martin (Eq. (1)) shares this same difficulty, for example, defect production and annihilation depend on the local compositions, phases and defect sinks. This difficulty appears to explain why alloys with nominal composition $< 3 \text{ at.}\% \text{ Cu(2)}$ failed to nucleate precipitates when initialized as a random solid solution, and why the effective temperature deduced from the SRO depends on microstructure. The description of the kinetic behavior also shows overall agreement with the METM, but it too suffers from the effects of an inhomogeneous microstructure. For example, precipitate dissolution during SPD behaves similarly to thermal dissolution for large precipitates, but not for small precipitates when precipitate cutting becomes important (see Fig. 6). Possibly this influence of inhomogeneities in the microstructure as well as various alloy properties could be included in a more general *effective free energy model*. For example, the work of Ravishanker et al. [46] considered the effect of compositional inhomogeneities on the local effective free energy; however, that approach would defeat the model's main purpose

of providing a general guide to phase evolution in alloy systems from simple considerations of the alloy thermodynamics and elastic properties.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.actamat.2021.116826](https://doi.org/10.1016/j.actamat.2021.116826).

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